

Recursive Estimation	
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1 Probability Theory	
1.1 Random Variables	
1.1.1 Discrete Random Variables	
DRV – Discrete Random Variable defined by $\mathcal{X}, p_x: \mathcal{X} \rightarrow \mathbb{R}$ $\mathcal{X} \subset \mathbb{Z}$ of all possible outcomes $p_x(\bar{x}) \geq 0$ and $\sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x}) = 1$ or $\sum_{\bar{x} \in \mathcal{X}} p_x(y \bar{x} \bar{y}) = 1$	
Margin. / Sum Rule	$p(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy}(\bar{x}, \bar{y})$
Cond. / Product Rule	$p_{x y}(\bar{x} \bar{y}) := \frac{p_{xy}(\bar{x}, \bar{y})}{p_y(\bar{y})}$
Total Prob. Theorem	$p_x(\bar{x}) := \sum_{\bar{y} \in \mathcal{Y}} p_{x y}(\bar{x} \bar{y})p_y(\bar{y})$
1.1.2 Continuous Random Variables	
CRV – Continuous Random Variable: DRV $\sum \rightsquigarrow$ CRV $\int$ - Set $\mathcal{X} \subseteq \mathbb{R}$ section of real line - PDF p_x satisfies $p_x(\bar{x}) \geq 0$ and $\int_{\mathcal{X}} p_x(\bar{x})d\bar{x} = 1$	
Caution: $Pr(x = \bar{x}) = 0, \quad Pr(x \in (a, b)) = \int_a^b p_x(\bar{x})d\bar{x}$	
1.1.3 Conditional PDF	
Conditional PDF	
Margin. / Sum Rule	$p(\bar{x} \bar{z}) = \sum_{\bar{y} \in \mathcal{Y}} p_{x y z}(\bar{x}, \bar{y} \bar{z})$
Cond. / Product Rule	$p_{x yz}(\bar{x} \bar{y}, \bar{z}) := \frac{p_{x y z}(\bar{x}, \bar{y} \bar{z})}{p_{y z}(\bar{y} \bar{z})}$
Independence	
$p(x y) = p(x) \Leftrightarrow p(y x) = p(y) \Leftrightarrow p(x, y) = p(x)p(y)$ $p(x, y, z) = p(x, y z)p(z) \rightarrow x, y$ indep. $p(x, y z) = p(z x, y)$	
Conditional Independence	
The knowledge of z makes x and y independent:	
$p(x y, z) = p(x z) \Leftrightarrow p(x, y z) = p(x z)p(y z)$	
Caution!!! in general we still have: $p(x, y) \neq p(x)p(y)$	
Caution!!! Independence $\nRightarrow$ Conditional Independence	
1.2 Expectation and Variance	
1.2.1 Expectation	
Definition: Integral for CRV	
$\mathbb{E}_x[x] = \sum_{\bar{x} \in \mathcal{X}} \bar{x}p_x(\bar{x})$	
- Linearity: $\mathbb{E}_{xy}[a + bx + cy] = a + b\mathbb{E}_x[x] + c\mathbb{E}_y[y]$ - Multi Variable: $\mathbb{E}_{xy}[g(x, y)] = \sum_{\bar{y}} \sum_{\bar{x}} g(\bar{x}, \bar{y})p_{xy}(\bar{x}, \bar{y})$ - Intependence of $x, y \Rightarrow \mathbb{E}_{xy}[xy] = \mathbb{E}_x[x]\mathbb{E}_y[y]$	
Law of Unconcious Statistician for $y = g(x)$	
$\mathbb{E}_y[y] = \sum_{\bar{y} \in \mathcal{Y}} \bar{y}p_y(\bar{y}) = \sum_{\bar{x} \in \mathcal{X}} g(\bar{x})p_x(\bar{x})$	
1.2.2 Variance (generally a matrix)	
$\text{Var}[x] = \mathbb{E} \left[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^\top \right] \stackrel{1D}{=} \mathbb{E} \left[x^2 \right] - \mathbb{E}[x]^2$	
Linearity	
$\text{Var}_x[a + bX + cY] = b^2 \text{Var}_X[x] + c^2 \text{Var}_Y[y] + 2bc \text{Cov}[X, Y]$	
Covariance: if independent $\text{Cov}(X, Y) = 0$	
$\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$	
1.3 Distribution Sampling	
CDF – Cumulative Distribution Function	
$\hat{F}_x(\bar{x}) := \sum_{\hat{x}=-\infty}^{\bar{x}} \hat{p}_x(\hat{x}) = Pr(x \leq \bar{x}) = \int_{-\infty}^{\bar{x}} p_x(\hat{x})d\hat{x}$	
A1 – 1 DRV $\bar{u}$ sample of $u \sim \mathcal{U}$, solve for $\bar{x}$ such that	
$\hat{F}_x(\bar{x} - 1) < \bar{u} \leq \hat{F}_x(\bar{x})$	
A2 – Multiple DRV or CRV	
Works for finite & infinite # of elements	
- Decompose $\hat{p}(\bar{x}, \bar{y}) = \hat{p}_{x y}(\bar{x} \bar{y})\hat{p}_y(\bar{y})$ - Apply A1 to get sample $\bar{y}$ via $\hat{p}_y(\bar{y})$ - With $\bar{y}$ fixed, apply A1 to get sample $\bar{x}$ via $\hat{p}_{x y}(\bar{x} \bar{y})$	
A3 – 1 CRV $\hat{p}_x$: pw-conti., bounded. $u \sim \mathcal{U}(0, 1)$ and solve:	
$\bar{u} = F_x(\bar{x}) \rightarrow \bar{x} = F_x^{-1}(\bar{u})$	

1.4 Change of Variables
DRV: $x = g(y)$
$p_x(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}: g(\bar{y}) = \bar{x}} p_y(\bar{y})$
CRV general case $y = g(x), \rightarrow x = g^{-1}(y)$:
$p_y(\bar{y}) = \left \frac{p_x(\bar{x})}{\frac{d g}{d x}(\bar{x})} \right , \quad \bar{x} = g^{-1}(y)$
CRV – Multivar. & Indep. Case: $z = x + y$
$p_z(\bar{z}) = \sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x})p_y(\bar{z} - \bar{x})$
CRV – Multivar. & Cond. Case: $z = g(w, x) \rightsquigarrow w = h(z, x)$
Assumptions: - $g(w, x), p_w$ cont. diff'bar - $g(w, x)$ strictly monotonic - For scalars $\det(x)$ is just x
$p_{z x}(\bar{z} \bar{x}) = p_{w x}(h(\bar{z}, \bar{x}) \bar{x}) \cdot \left \det \left(\frac{\partial}{\partial w} g(h(\bar{z}, \bar{x}), \bar{x}) \right) \right ^{-1}$
CRV – Multivar. & Cond. Case: $z = g(w) \rightsquigarrow w = h(z)$
$p_z(\bar{z}) = p_w(h(\bar{z})) \cdot \left \det \left(\frac{\partial}{\partial w} g(w) \Big _{w=h(\bar{z})} \right) \right ^{-1}$
1.5 Miscellaneous
Indep. multiplication: $z = xy \rightarrow p(z) = \sum_Y p_x(\frac{z}{y})p_y(y)$
Uniform Distribution
$p(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases} \quad \left \quad \begin{array}{l} \mathbb{E}\{x\} = \frac{a+b}{2} \\ \text{Var}\{x\} = \frac{(b-a)^2}{12} \end{array} \right.$
Triangular Distribution
$p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \leq x \leq b \\ 0 & \text{else} \end{cases}$
$F(x) = \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \leq x \leq c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \leq x \leq b \\ 1 & x > b \end{cases}$
$\mathbb{E}\{x\} = \frac{a+b+c}{3}, \quad \text{Var}\{x\} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$
Gaussian Distribution $\mathbf{x} = (x_1, \dots, x_D) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
$p_{\mathbf{x}}(\bar{\mathbf{x}}) = \frac{1}{(2\pi)^{D/2} \det(\boldsymbol{\Sigma})^{1/2}} \exp \left(-\frac{(\bar{\mathbf{x}} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu})}{2} \right)$
$\rightsquigarrow$ fully characterized by mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and Variance $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$
Affine transformation of Gaussian: $x \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
$\mathbf{y} = M\mathbf{x} + \mathbf{b} \rightarrow \mathbf{y} \sim \mathcal{N}(M\boldsymbol{\mu} + \mathbf{b}, M\boldsymbol{\Sigma}M^T)$
Linear combination of jointly GRV: $\mathbf{x}, \mathbf{y}$
$\mathbf{z} = M_x\mathbf{x} + M_y\mathbf{y} \Rightarrow \mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$
$\boldsymbol{\mu}_z = M_x\boldsymbol{\mu}_x + M_y\boldsymbol{\mu}_y$
$\boldsymbol{\Sigma}_z = M_x\boldsymbol{\Sigma}_xM_x^T + M_x\boldsymbol{\Sigma}_{xy}M_y + M_y\boldsymbol{\Sigma}_{xy}^T M_x + M_y\boldsymbol{\Sigma}_yM_y^T$
(Red symbols are zero if $\mathbf{x}, \mathbf{y}$ are independent)
Diagonal Variance $\rightsquigarrow y_i$ mut.indep. iff $\boldsymbol{\Sigma}$ diagonal
$p(y) = \prod_{i=1}^D \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left(-\frac{(y_i - \mu_i)^2}{2\sigma_i^2} \right)$
Scalar Gaussian Distribution $D = 1, \sigma^2 = \det(\boldsymbol{\Sigma}) \rightarrow \det(\boldsymbol{\Sigma})^{1/2} = \sigma$
$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right) \quad \left \quad \begin{array}{l} \mathbb{E}\{x\} = \mu \\ \text{Var}\{x\} = \sigma^2 \end{array} \right.$

2 Bayesian Tracking
2.1 Bayes Theorem
$p(x y) = \frac{p(y x)p(x)}{p(y)}$
x =system state, $p(x)$ =prior belief of state, z =observations related to state $p(z x)$ =observation likelihood, $p(x z)$ =posterior belief of state $p(z)$ =normalization constant $= \sum_x p(z x)p(x)$
For multiple observations:
$p(x z_1, \dots, z_N) = \frac{\overbrace{p(x)}^{\text{prior}} \prod_i \overbrace{p(z_i x)}^{\text{observation likelihood}}}{\underbrace{\sum_{x \in \mathcal{X}} p(x) \prod_i p(z_i x)}_{\text{normalization}}}$
2.2 Bayesian Tracking
Model
Objective: Calculate pdf of state $p(x(k) z(1:k))$
$x(k) = q_{k-1}(x(k-1), v(k-1)) \in \mathcal{X}$
$z(k) = h_k(x(k), w(k))$
Prior Update - State Prediction
$p(x(k) z(1:k-1)) = \sum_{x(k-1)} \underbrace{p(x(k) x(k-1))}_{\text{process model}} \underbrace{p(x(k-1) z(1:k-1))}_{\text{previous iteration}}$
Measurement Update
$p(\bar{x}(k) \bar{z}(1:k)) = \frac{\underbrace{p(\bar{z}(k) \bar{x}(k))}_{\text{measurement model}} \underbrace{p(\bar{x}(k) \bar{z}(1:k-1))}_{\text{prior}}}{\underbrace{\sum_{i \in \mathcal{X}} p(\bar{z}(k) i)p(i \bar{z}(1:k-1))}_{\text{normalization}}}$
Computer Implementation
1. Enumerate state space $\mathcal{X} = \{0, 1, \dots, N-1\}$
2. Posterior define $\mathbf{a}_{k k}^i := p(x(k) z(1:k))(i \bar{z}(1:k))$
3. Prior define $\mathbf{a}_{k k-1}^i := p(x(k) z(1:k-1))(i \bar{z}(1:k-1))$
Initialization $\mathbf{a}_{0 0}^i = p_{x(0)}(i) \quad i = 0, \dots, N-1$
$\left. \begin{array}{l} \mathbf{a}_{k k-1}^i = \sum_{j=0}^{N-1} p_{x(k) x(k-1)}(i j)\mathbf{a}_{k-1 k-1}^j \\ \mathbf{a}_{k k}^i = \frac{p_{z(k) x(k)}(\bar{z}(k) i)\mathbf{a}_{k k-1}^i}{\sum_{j=0}^{N-1} p_{z(k) x(k)}(\bar{z}(k) j)\mathbf{a}_{k k-1}^j} \end{array} \right\} \text{Iterate } k > 0$
3 Extracting Estimates
3.1 General Measurement
$z = Hx + w, \quad z \in \mathbb{R}^m, x \in \mathbb{R}^n, H \in \mathbb{R}^{m \times n}, w \in \mathbb{R}^m$ m Measurements, n States, $w_i \sim \mathcal{N}(0, R), x, w, i$ assumed mut. indep.
3.2 Likelihood Estimators
3.2.1 ML-Maximum Likelihood
Choose x to make $\bar{z}$ most likely:
$\left. \begin{array}{ll} x \in \mathcal{X} & \text{unknown} \\ p_x(\bar{x}) & \text{unknown} \end{array} \right\} \hat{x}^{ML} := \underset{\bar{x} \in \mathcal{X}}{\operatorname{argmax}} \left[p_{z x}(\bar{z} \bar{x}) \right]$
$p_{z x}(\bar{z} \bar{x})$ is var. – technically not a PDF
3.2.2 MAP-Maximum A Posteriori
ML given observation & prior belief about x :
$\left. \begin{array}{ll} x \in \mathcal{X} & \text{unknown} \\ p_x(\bar{x}) & \text{known} \end{array} \right\} \hat{x}^{MAP} := \underset{\bar{x} \in \mathcal{X}}{\operatorname{argmax}} \left[p_{z x}(\bar{z} \bar{x})p_x(\bar{x}) \right]$
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$
3.3 Least Squares Estimators
3.3.1 Linear Least Squares
$\hat{x}^{LS} := \underset{x}{\operatorname{argmin}} \left[(z - Hx)^\top (z - Hx) \right] = (H^\top H)^{-1} H^\top z$
$\hat{x}^{LS} = \hat{x}^{ML}$, when errors are indep. 0-mean, same variance, $\mathcal{N}$ -distributed
3.3.2 Weighted Least Squares
$\hat{x}^{WLS} := \underset{x}{\operatorname{argmin}}_x \left[(\bar{z} - H\hat{x})^\top R^{-1} (\bar{z} - H\hat{x}) \right]$
$= (H^\top R^{-1} H)^{-1} H^\top R^{-1} \bar{z}$
Incorporate Prior Knowledge on $\mathbf{x}$:
$r \sim \mathcal{N}(0, P_x) \Rightarrow x := \bar{x} + r, \mathbb{E}\{x\} = \hat{x}_0, \text{Var}\{x\} = P_x$
Introduce Extended System:
$\tilde{z} = \begin{bmatrix} \hat{x}_0 \\ z(1:k) \end{bmatrix}, \tilde{H} = \begin{bmatrix} \mathbb{I} \\ H(1:k) \end{bmatrix} \tilde{w} = \begin{bmatrix} -r \\ w(1:k) \end{bmatrix}, \tilde{R} = \begin{bmatrix} P_x & 0 \\ 0 & R(1:k) \end{bmatrix}$

3.3.3 Recursive Least Squares

Interpretation Update $\hat{x}$ recursively when new $\bar{z}(k)$ arrives
Estimation Error: $e(k) = (\mathbb{I} - KH) e(k-1) - Kw(k)$
Objective Minimize MSE with $P(k) = \text{Var} \{e(k)\}$

$$J(k) := \mathbb{E} \left\{ e^\top(k) e(k) \right\} = \mathbb{E} \{ \text{trace}(P(k)) \}$$

Algorithm

Initialization: $\hat{x}(0) = \hat{x}_0, \quad P(0) = P_x = \text{Var}\{x\}$

Recursion: Observe $\bar{z}(k)$, then update:

$$K(k) = P(k-1)H(k)^\top \left(H(k)P(k-1)H(k)^\top + R(k) \right)^{-1}$$

$$\hat{x}(k) = \hat{x}(k-1) + K(k) (\bar{z}(k) - H(k)\hat{x}(k-1))$$

$$P(k) = (\mathbb{I} - K(k)H(k))P(k-1)(\mathbb{I} - K(k)H(k))^\top + K(k)R(k)K(k)^\top$$

4 Kalman Filter

4.1 Kalman Filter

Model (Linear Time-Invariant)

$$x(k) = A(k-1)x(k-1) + u(k-1) + v(k-1)$$

$$z(k) = H(k)x(k) + w(k)$$

$x(0), v(\cdot), w(\cdot)$ ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$

Auxiliary Variables

Problem Formulation want $p_{x(k)}|z(1:k) \rightsquigarrow$ bayesian tracking provides this
BUT we have CRVs and don't want integrals $\Rightarrow$ exploit linearity

Auxiliary Variables:

Init: $x_m(0) = x(0)$

S1: $x_p(k) = A(k-1)x_m(k-1) + u(k-1) + v(k-1)$

S2: $z_m(k) = H(k)x_p(k) + w(k)$

$$p_{x_m(k)}(\xi) := p_{x_p(k)|z_m(k)}(\xi|\bar{z}(k)) \quad \forall \xi$$

Equivalency:

$$p_{x_p(k)}(\xi) = p_{x(k)|z(1:k-1)}(\xi|\bar{z}(1:k-1))$$

$$p_{x_m(k)}(\xi) = p_{x(k)|z(1:k)}(\xi|\bar{z}(1:k))$$

GRV x_p and x_m are GRVs for $k = 1, 2, \dots$

$$\hat{x}_p(k) := \mathbb{E}[x_p(k)] \quad P_p(k) := \text{Var}[x_p(k)]$$

$$\hat{x}_m(k) := \mathbb{E}[x_m(k)] \quad P_m(k) := \text{Var}[x_m(k)]$$

Recursive Update Equations

Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step

$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + Q(k-1)$$

S2 – A Posteriori Update / Measurement Update

$$P_m(k) = \left(P_p^{-1}(k) + H^\top(k)R^{-1}(k)H(k) \right)^{-1}$$

$$\hat{x}_m(k) = \hat{x}_p(k) + \underbrace{P_m(k)H^\top(k)}_{\text{correctionfactor}} \underbrace{R^{-1}(k)}_{\text{predictionerror}} (\bar{z}(k) - H(k)\hat{x}_p(k))$$

Alternative Equations – all matrices/vectors at time k

$$K(k) = P_p(k)H^\top(k) \left(H(k)P_p(k)H^\top(k) + R(k) \right)^{-1}$$

S2 – Kalman Filter Gain:

$$\hat{x}_m = \hat{x}_p + K(\bar{z} - H\hat{x}_p)$$

$$P_m = \underbrace{(\mathbb{I} - KH)P_p}_{\text{cheaper}} = \underbrace{(\mathbb{I} - KH)P_p(\mathbb{I} - KH)^\top + KRK^\top}_{\text{less sensitive to num. errors}}$$

KF Remarks

- The KF is the analytical solution to the Bayesian state estimation problem for linear systems with Gaussian distributions. (Best You can do)
- Otherwise, the KF is no longer optimal. It is the best linear estimator in the MMSE sense but nonlinear might do better.
- $A(k), H(k), Q(k), R(k) \quad \forall k$ as well as $P_0 \rightsquigarrow$ KF matrices $P_p(k), P_m(k), K(k), K(k)$ can be computed offline, K no f() of Measurm.
- KF is linear and time-varying:

$$\hat{x}_m(k) = \hat{A}(k) \cdot \hat{x}_m(k-1) + \hat{B}(k) \cdot u(k-1) + K\bar{z}(k)$$

$$\hat{A}(k) = (\mathbb{I} - K(k)H(k))A(k-1), \quad \hat{B}(k) = (\mathbb{I} - K(k)H(k))$$
 - Among class of linear, unbiased estimators, the KF minimizes MSE
 - Not all system matrices need to be pd. some can also be psd. as long as inversions are possible in the update equations.

4.1.1 Detectability / Stabilizability
Detectability Conditions
Pair (A, H) detectable
$\Leftrightarrow$ For deterministic LTI Sys. $(x(k)=Ax(k-1), z(k)=Hx(k))$ $\lim_{k \rightarrow \infty} z(k) = 0 \Rightarrow \lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$
$\Leftrightarrow \begin{bmatrix} A - \lambda \mathbb{I} & B \\ H & \end{bmatrix}$ is full rank for all $\lambda \in \mathbb{C}$ with $ \lambda \geq 1$ (PBH-Test)
$\Leftrightarrow$ Eigenvalues of $A - LH$ (or $(\mathbb{I} - LH)A$) can be placed within unit circle by suitable choice of $L \in \mathbb{R}^{n \times m}$
Stabilizability Conditions
Pair (A, B) stabilizable
$\Leftrightarrow$ For deterministic LTI Sys. $(x(k)=Ax(k-1)+Bu(k-1))$ $\exists u(0 : k-1)$ s.t. $\lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$
$\Leftrightarrow [A - \lambda \mathbb{I} \ B]$ full rank $\forall \lambda \in \mathbb{C}$ with $ \lambda \geq 1$ (PBH-Test)
$\Leftrightarrow$ Eigenvalues of $A - BK$ (or $(\mathbb{I} - BK)A$) can be placed within unit circle by suitable choice of $K \in \mathbb{R}^{m \times n}$
Observability / Reachability
if $(A^\top, B^\top)$ detectable
Observability Matrix $O = \begin{bmatrix} H \\ HA^{n-1} \end{bmatrix}$
implies detectability
Reachability Matrix $R = \begin{bmatrix} A & A^{n-1}B \end{bmatrix}$
implies stabilizability
4.1.2 Asymptotic Properties of KF
Model assume TI System, stationary Distributions (A, H, Q, R) constants) $\rightsquigarrow$ the KF is still time-varying! Change of Variance
$P_p(k+1) = \left(AP_p(k)A^\top \right) + Q$
$ - \left(AP_p(k)H^\top \right) \left(HP_p(k)H^\top + R \right)^{-1} \left(HP_p(k)A^\top \right)$
Scalar Case $P_p(k+1) = \frac{a^2 r P_p(k)}{h^2 P_p(k) + r} + q \Leftarrow f(P_p(k))$
Summary <i>Variance Converges to unique steady-state sol'n provided either $a < 1$ or if $a \geq 1 \Rightarrow h \neq 0$ and $q > 0$</i>
4.1.3 Steady-State KF
DARE – Discrete Algebraic Ricatti Equation (P_∞ is P_p)
$P_\infty = AP_\infty A^\top + Q - AP_\infty H^\top (HP_\infty H^\top + R)^{-1} HP_\infty A^\top$
Steady-State KF Gain: $K_\infty = P_\infty H^\top (HP_\infty H^\top + R)^{-1}$
Steady State KF Equations $\hat{x}(k) := \hat{x}_m(k)$
$\hat{x}(k) = (\mathbb{I} - K_\infty H)A\hat{x}(k-1) + (\mathbb{I} - K_\infty H)u(k-1) + K_\infty \bar{z}(k)$
$\hat{x}(k) = \hat{A} \cdot \hat{x}(k-1) + \hat{B} \cdot u(k-1) + K_\infty \cdot \bar{z}(k), \quad \hat{x}(0) = x_0$
Error dynamics (also holds for non ss-KF) $e(k) = x(k) - x_f(k)$
$e(k) = (\mathbb{I} - K_\infty H)Ae(k-1) + (\mathbb{I} - K_\infty H)v(k-1) - K_\infty w(k)$
$\mathbb{E}[e(k)] = (\mathbb{I} - K_\infty H)A \mathbb{E}[e(k-1)]$
If $\mathbb{E}[e(0)] = 0$ then $\mathbb{E}[e(k)] = 0 \quad \forall k \rightsquigarrow$ KF is unbiased
$\mathbb{E}[e(k)] \xrightarrow{k \rightarrow \infty} 0 \Leftrightarrow (\mathbb{I} - K_\infty H)A$ stable (all $\lambda < 1$)
Theorem for $R > 0, Q \geq 0, G$ s.t. $Q = GG^\top$
$\Leftrightarrow (A, H)$ is detectable and (A, G) is stabilizable
$\Leftrightarrow$ DARE has unique pos. semidef. sol. $P_\infty \geq 0$, resulting $(\mathbb{I} - K_\infty H)A$ is stable and $\lim_{k \rightarrow \infty} P_p(k) = P_\infty \quad \forall P_p(1) \geq 0$ and $\forall P_m(0) = P_0 \geq 0$
(A,H) Detectable: can observe all unstable modes
(A,G) Stabilizable: noise excites unstable modes
Note if $Q > 0 \Rightarrow (A, G)$ always stabilizable If $Q = 0$ multiple valid solutions can be possible
4.2 Extended Kalman Filter
$x(k) = q_{k-1}(x(k-1), v(k-1))$
$z(k) = h_k(x(k), w(k)) \quad x(0), v(\cdot), w(\cdot)$ mutually ind.
$\mathbb{E}[x(0)] = x_0, \text{Var}[x(0)] = P_0 \quad \quad \mathbb{E}[w(k)] = 0, \text{Var}[w(k)] = R(k)$
$\mathbb{E}[v(k-1)] = 0, \text{Var}[v(k-1)] = Q(k-1)$

EKF Equations
Initialization: $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step
$\hat{x}_p(k) = q_{k-1}(\hat{x}_m(k-1), 0)$
$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + L(k-1)Q(k-1)L^\top(k-1)$
$A(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial x}, \quad L(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial v}$
S2 – A Posteriori Update / Measurement Update (at k)
$K = P_p H^\top \left(H P_p H^\top + M R M^\top \right)^{-1}$
$\hat{x}_m = \hat{x}_p + K \left(\bar{z} - h_k(\hat{x}_p(k), 0) \right)$
$P_m = (\mathbb{I} - K(k)H(k)) P_p(k)$
$H(k) := \left. \frac{\partial h_k(x, w)}{\partial x} \right _{x=\hat{x}_p(k), w=0}, \quad M(k) := \left. \frac{\partial h_k(x, w)}{\partial w} \right _{x=\hat{x}_p(k), w=0}$
Caution
- Process & measurement noise assumed 0-mean A, L, H, M lin. about current $\hat{x}_m \rightsquigarrow$ cannot compute offline - Prior update only accurate if $\mathbb{E}\{\cdot\}$ & $q_{k-1}(\cdot)$ commute EKF Variables DO NOT capture true conditional mean & var.
4.2.1 Hybrid EKF
System Process – non-lin, CT, Measurement – non-lin, DT, $x[k] := x(kT)$
$\dot{x}(k) = q(x(t), v(t), t) \mathbb{E}[x(0)] = x_0, \text{Var}[x(0)] = P_0$
$\mathbb{E}[v(t)] = 0, \text{Var}[v(t)v(t+\tau)^\top] = Q_c \delta(\tau)$
$z[k] = h_k(x[k], w[k]) \quad \mathbb{E}\{w[k]\} = 0, \text{Var}[w[k]] = R$
CT White Noise $\delta(t)$ = Dirac pulse
$\mathbb{E}\{v(t)\} = 0$ and $\mathbb{E}\{v(t)v^\top(t+\tau)\} = Q_c \delta(\tau)$
<i>True CT White noise does not exist $\rightarrow$ Would require infinite power</i>
Hybrid EKF Equations
Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step (Solve ODEs)
Solve, for $(k-1)T \leq t \leq kT$ and $\hat{x}((k-1)T) = \hat{x}_m[k-1]$
$\dot{\hat{x}}(t) = q(\hat{x}(t), 0, t) \Rightarrow \hat{x}_p[k] := \hat{x}(kT)$
Solve, for $(k-1)T \leq t \leq kT$ and $P((k-1)T) = P_m[k-1]$
$\dot{P}(t) = AP + PA^\top + LQ_c L^\top \Rightarrow P_p[k] := P(kT)$
$A(t) := \frac{\partial q}{\partial x}(\hat{x}(t), 0, t), \quad L(t) := \frac{\partial q}{\partial v}(\hat{x}(t), 0, t)$
S2 – A Posteriori Update / Measurement Update
<i>Measurement model still DT – S2 identical to standard EKF</i>
Caution
- Process & measurement noise assumed 0-mean - ODEs typically solved numerically: Accuracy $\uparrow$ = Comp. Cost $\uparrow$
5 Particle Filter
5.1 Monte Carlo Sampling
Samples N DRV's y^n of samples of $y \in \mathcal{Y} = \{1, \dots, \bar{Y}\} \sim p_y$
Helper Variables
$s_i^n = \delta(i - y^n) = \begin{cases} 1 & \text{if } y^n = i \\ 0 & \text{otherwise} \end{cases} \quad \left \begin{array}{l} i = 1, \dots, \bar{Y} \\ n = 1, \dots, N \end{array} \right.$
Average – s_i converges to $p_y(i)$ by Law of Large Numbers
$\frac{1}{N} \sum_{n=1}^N s_i^n \xrightarrow{N \rightarrow \infty} \mathbb{E}\{s_i^n\} = p_y(i) \quad p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \bar{s}_i^n$
Approximation
For $y \in \mathcal{Y}$, it holds $p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \delta(i - \bar{y}^n), \quad i \in \mathcal{Y}$
For $x = g(y)$, with $x \in \mathcal{X} := g(\mathcal{Y})$, it holds
$p_x(j) \approx \frac{1}{N} \sum_{n=1}^N \delta(j - g(\bar{y}^n)), \quad j \in g(\mathcal{Y})$
Monte Carlo Sampling CRV
Helper Variables
$s_a^n = \int_a^{a+\Delta y} \delta(i - y^n) d\xi = \begin{cases} 1 & \text{if } a \leq y^n < a + \Delta y \\ 0 & \text{otherwise} \end{cases}$
Average – s_a converges to $p_y(\xi)$ by Law of Large Numbers
$\int_a^{a+\Delta y} \frac{1}{N} \sum_{n=1}^N \delta(\xi - y^n) d\xi \xrightarrow{N \rightarrow \infty} \int_a^{a+\Delta y} p_y(\xi) d\xi$

Approximation
For $y \in \mathcal{Y}$, it holds $p_y(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{y}^n) \forall \xi$
For $x = g(y)$, it holds $p_x(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - g(\bar{y}^n)) \forall \xi$
5.2 Modeling
System <i>Non-linear, discrete-time:</i>
$x(k) = q_{k-1}(x(k-1), v(k-1))$
$z(k) = h_k(x(k), w(k))$
$x(0), \{v\}, \{w\}$ mutually indep. can be D- or CRVs, known PDFs
Objective
Bayesian State Estimator with Monte Carlo Sampling
Init: $x_m(0) \quad := \quad x(0)$ S1: $x_p(k) \quad := \quad q_{k-1}(x_m(k-1), v(k-1))$ S2: $z_m(k) \quad := \quad h_k(x_p(k), w(k))$ $p_{x_m(k)}(\xi) \quad := \quad p_{x_p(k) z_m(k)}(\xi \bar{z}(k)) \quad \forall \xi$
Similar proof for $p_{x_p(k)}(\xi), p_{x_m(k)}(\xi) \forall \xi$ as for KF aux. var.
5.3 Particle Filter - Equations
Initialization Draw N samples $\{\bar{x}_m^n(0)\}$ from $p_x(0)$
S1 – Prior Update / Prediction Step
Apply process equation to the particles $\{\bar{x}_m^n(k-1)\}$
$\bar{x}_p^n(k) := q_{k-1}(\bar{x}_m^n(k-1), \bar{v}^n(k-1)), \quad n = 1, \dots, N$
Requires N noise samples from $p_v(k-1)$
S2 – A Posteriori Update / Measurement Update
Scale each particle by measurement likelihood:
$\beta_n = \alpha p_{z(k) x(k)}(\bar{z}(k) \bar{x}_p^n(k)), \quad n = 1, \dots, N$
with α normalization s.t $\sum_{n=1}^N \beta_n = 1$
$\alpha = \left(\sum_{n=1}^N p_{z_m(k) x_p(k)}(\bar{z}(k) \bar{x}_p^n(k)) \right)^{-1}$
Resample to get N posterior particles $\{\bar{x}_m^n(k)\}$, w/ eq. weights
Repeat N times: - Select random number r uniformly on $(0, 1)$ - Pick particle $\bar{n}$ s.t $\sum_{n=1}^{\bar{n}-1} \beta_n < r \leq \sum_{n=1}^{\bar{n}} \beta_n$
New particles all have equal weight
$p_{x_m(k)}(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{x}_m^n(k)), \quad \forall \xi$
Intuition
Process Update – Propagate N particles through system dynamics. Provided N large, $\{x_m^n(k-1)\} \approx p_{x_m(k-1)} \rightsquigarrow \{\bar{x}_p^n(k)\} \approx p_{x_p(k)}$
Measurement Update – treat each particle separately. At points of high prior, we have many particles. Posterior has same particles, but scaled by measurement likelihood.
5.4 Roughening
Sample Impoverishment <i>Lumpiness</i> – when we resample, only retain subset of particles. For finite N this is problem.
Roughening
Perturb particles after resampling
$\bar{x}_m^n(k) \leftarrow \bar{x}_m^n(k) + \Delta x^n(k)$
$\Delta x^n(k)$ drawn from 0-mean, finite-var. distribution
Example Method for Δx^n: σ_i stand. dev. of $\Delta x_i^n(k)$
$\sigma_i = KE_i N^{-\frac{1}{d}}$
K: tuning parameter, typically $K \ll 1$
d: dimension of state space
E_i : max inter-sample variability $\max_{n_1, n_2} \bar{x}_{m,i}^{n_1} - \bar{x}_{m,i}^{n_2} $
$N^{-\frac{1}{d}}$: node spacing on grid

6 Observer Based Control
6.1 LTI Observer
LTI System
$x(k) = Ax(k-1) + Bu(k-1) + v(k-1)$
$z(k) = Hx(k) + w(k)$
v, w 0-mean noise CRVs
$\mathbb{E}\{\hat{x}(k)\} \xrightarrow{k \rightarrow \infty} \mathbb{E}\{x(k)\}, \text{Var}\{\hat{x}(k)\}$ bounded
Luenberger Observer
Same structure as steady-state KF:
$\hat{x}(k) = A\hat{x}(k-1) + Bu(k-1) + K(\bar{z}(k) - \hat{z}(k))$
$\phantom{\hat{x}(k)} = (\mathbb{I} - K_\infty H)A\hat{x}(k-1)$
$\phantom{\hat{x}(k)} + (\mathbb{I} - K_\infty H)Bu(k-1) + K_\infty \bar{z}(k)$
$\hat{z}(k) = H(A\hat{x}(k-1) + Bu(k-1))$
Error $e(k) = x(k) - \hat{x}(k)$
in absence of noise $\bar{z}(k) = z(k)$
$e(k) = (\mathbb{I} - KH)Ae(k-1)$
$(\mathbb{I} - KH)A$ must be stable (all $\lambda < 1$) for $e \xrightarrow{k \rightarrow \infty} 0$ - Such K exists iff (A, H, A) detectable $\Leftrightarrow (A, H)$ detectable
Observer Design
If (A, H) detectable, construct K
Pole Placement: place λ of e -dynamics of observer modes at desired locations
Steady State KF: Yields optimal K (min steady-state MSE) given statistics of v, w . Also need (A, G) stabilizable, $\text{Var}\{v(k)\} = GG^\top$
Alternative Formulation
$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + K(\bar{z}(k) - \hat{z}(k))$
$\hat{z}(k) = H\hat{x}(k) \rightarrow$ delayed measurement
Error Dynamics
$e(k+1) = (A - KH)e(k)$
6.2 Static State Feedback Control
System Model
LTI, no noise, deterministic, perf. state info $z(k) = x(k)$:
Feedback Law $u(k) = Fx(k) = Fz(k)$ CL Dynamics $x(k) = (A + BF)x(k-1)$
$(A + BF)$ must be stable (all $\lambda < 1$) - Such F exists $\Leftrightarrow (A, B)$ stabilizable
Controller Design
Pole Placement: place (controllable) poles at des. CL loc.
LQR: find F that min. quad. cost $(\bar{Q} = \bar{Q}^\top \geq 0, \bar{R} = \bar{R}^\top \geq 0)$
$J_{\text{LQR}} = \sum_{k=0}^\infty x^\top(k)\bar{Q}x(k) + u^\top(k)\bar{R}u(k)$
LQR Design
Assuming (A, B) stabil., (A, G) detect.
$F = -(B^\top PB + \bar{R})^{-1} B^\top PA$
DARE
Same as KF but $A \rightarrow A^\top, H \rightarrow B^\top, Q \rightarrow \bar{Q}, R \rightarrow \bar{R}$
$P = A^\top PA + \bar{Q} - A^\top PB(B^\top PB + \bar{R})^{-1} B^\top PA$
6.3 Seperation Principle
Stable Observer + Controller $\rightsquigarrow u(k) = F\hat{x}(k)$
Dynamics
$\begin{bmatrix} x(k) \\ e(k) \end{bmatrix} = \begin{bmatrix} A + BF & -BF \\ 0 & (\mathbb{I} - KH)A \end{bmatrix} \begin{bmatrix} x(k-1) \\ e(k-1) \end{bmatrix}$
Separation Principle
$(\mathbb{I} - KH)A$ and $(A + BF)$ stable $\rightsquigarrow$ Overall system stable - Still stable with noise, Generalizable to time-varying case - Does NOT hold for nonlinear systems
Separation Theorem
Optimal strategy for control problem
$J_{\text{LQG}} = \lim_{N \rightarrow \infty} \mathbb{E} \left\{ \frac{1}{N} \sum_{k=0}^{N-1} \left[x^\top(k)\bar{Q}x(k) + u^\top(k)\bar{R}u(k) \right] \right\}$
- Design steady state KF to provide $\hat{x}(k)$ (no dependency on $\bar{Q}, \bar{R}$) - Design optimal state-FB law $u(k) = Fx(k)$ for determ. LQR problem minimizes J_{LQR} w/o noise (no dependency on noise statistics Q, R) - Put together